



Compare code. Coach with clarity.

A practical guide to comparing a student submission with an instructor reference and turning differences into specific learning steps.

PURPOSE

CodeMirror is a lightweight coaching aid. It puts two solutions side by side, points to changed lines, and gives the instructor a useful starting point for a focused conversation.

How it works

1. Add both solutions

Paste the student submission in the left pane and the instructor reference in the right pane. Select a language or leave Auto detect selected.

2. Compare the text

The app normalizes whitespace, then matches lines and groups additions and removals. The code is compared as text; it is never executed.

3. Read the coaching notes

Each changed area shows the student version beside the reference version, with a short explanation of why the difference may matter.

4. Choose a practice step

Use the suggested prompt to discuss the control flow, edge cases, testing, naming, formatting, or implementation. Recheck the assignment before judging correctness.

What a finding looks like

STUDENT VERSION

```
return total / len(numbers)
```

REFERENCE VERSION

```
if not numbers: raise ValueError(...)
```

COACHING PROMPT

What should happen for an empty list? Try that edge case and explain the behavior you expect.



Benefits for teaching

Make feedback concrete, consistent, and easier for students to act on.

See the exact change Grouped differences and paired snippets keep the conversation anchored to the code in front of you.	Spend more time teaching A quick first pass surfaces likely review areas, so instructors can focus their time on explanation and student reasoning.
Give a next step Each finding includes a plain-language explanation and a practice prompt, not only a raw diff.	Support independent learning Students can compare their approach with the reference, investigate an edge case, and return with a reasoned question.
Keep submissions local The app runs in a browser tab and sends no source code to a server. It does not save submissions after the page closes.	Use with many languages The text comparison is language-agnostic; feedback categories use lightweight pattern matching and are only hints.

Reading the learning snapshot

The similarity percentage summarizes matching lines after whitespace normalization. The category bars are heuristic indicators derived from the number and type of differences. They are not a grade, test result, or proof that either solution is correct. Two different implementations can both satisfy the same requirements.

A good instructor workflow

Check the assignment requirements first. Ask the student to explain the approach, trace an ordinary input, then try an edge case. Use the reference as a discussion aid, and accept valid alternatives when they meet the learning goals.

Privacy and limitations

- **Local processing:** code comparison happens in the browser; there is no account, server, or upload.
- **No code execution:** the app cannot run submissions, inspect runtime errors, or verify outputs.
- **Pattern-based feedback:** language cues are lightweight heuristics and may misclassify code.
- **Reference is not the only answer:** the reference may show one preferred approach. Review student intent and task criteria before drawing conclusions.

[Open index.html to start.](#) [Paste both solutions, compare, then use the prompts to guide an instructor-led review.](#)